

Is Microbial Keystone Species Identification Invariant to Model Inductive Bias?

William Little

WLITTLE@HAWAII.EDU

*Department of Information and Computer Sciences
University of Hawai'i at Mānoa
Honolulu, HI 96822, USA*

Editor:

Abstract

Identifying keystone species within microbial communities remains a major challenge in computational biology due to the complexity of microbial interactions and the difficulty of experimental perturbation studies. In 2024, Wang et al. introduced the Data-driven Keystone-species Identification (DKI) framework, which used the cNODE2 Neural ODE architecture to infer keystone species from learned mappings between binary species presence vectors and relative abundance compositions. Although the framework achieved strong predictive performance, it remains unclear whether the inferred keystone species reflect stable ecological signals or artifacts of the underlying model architecture. In this work, we investigate the robustness of the DKI framework across multiple predictive architectures with different inductive biases. Despite differences in predictive performance and perturbation magnitude, the inferred keystone species rankings remained highly consistent across models, suggesting that DKI-style keystone identification may be more robust to model inductive bias than previously understood.

1 Introduction

Recent advances in deep learning have motivated the development of data-driven approaches for modeling microbial community dynamics and identifying influential species. In 2024, Wang et al. [7] proposed the Data-driven Keystone-species Identification (DKI) framework, as the first fully data-driven framework for microbial keystone species identification. Their approach used the Neural ODE [2] architecture to learn a mapping from binary species presence vectors to relative abundance compositions, allowing for simulated species removal experiments directly within the learned model. Keystone species are then identified by measuring the magnitude of predicted community change after species perturbations.

Although the DKI framework had strong predictive performance, it remains unclear whether the inferred keystone species reflect real biological signals or are artifacts of the underlying model architecture and inductive bias. Since the original study evaluated only the cNODE2 Neural ODE model, it is unknown if alternative predictive architectures would produce similar keystone species rankings after learning the same abundance prediction mapping.

In this work, four distinct predictive models with comparable predictive performance are trained on the same microbial abundance prediction task and compare the resulting keystone species rankings across models. If keystone inference is robust to architecture

choice, then the identified keystone rankings should remain highly consistent across models despite their different inductive biases.

2 Background

A *keystone species* is a species whose removal causes a disproportionately large effect on the structure and stability of an ecosystem relative to its abundance [5]. In microbial communities, keystone species can strongly influence biodiversity and overall community composition. Identifying microbial keystone species experimentally is challenging due to the complexity of microbial ecosystems and the difficulty of performing controlled perturbation experiments.

The method proposed in [7] trains a predictive model to learn a mapping from a binary species presence vector z to a relative abundance composition p . Keystone species are identified from a thought experiment by simulating perturbation experiments in which individual species are removed from the community and the change in community composition is measured.

The DKI framework quantifies community change using the Bray-Curtis dissimilarity metric [1] from ecology:

$$BC(p, q) = \frac{\sum_i |p_i - q_i|}{\sum_i |p_i + q_i|}$$

where p represents the original microbial composition and q represents the predicted composition after species removal. Species whose removal produces large Bray-Curtis shifts are considered to have high keystone-ness. Global keystone-ness rankings are obtained by aggregating perturbation effects from all measured samples of the microbial community.

A key concept in this work is *inductive bias*, which refers to the assumptions a machine learning model makes about the structure of the data [3]. Different neural architectures have different inductive biases regarding how microbial communities evolve and interact. To investigate the robustness of DKI to model inductive bias, we evaluate several predictive architectures including linear models, residual multilayer perceptrons (Residual MLPs) [4], attention-based models [6], Neural Ordinary Differential Equations (NODEs), and the original cNODE2 architecture. Each of their inductive biases are explained in Table 1.

Neural ODEs [2] model hidden representations as continuous dynamical systems governed by differential equations:

$$\frac{dy}{dt} = f_\theta(y, t)$$

where f_θ is parameterized by a neural network. The cNODE2 architecture is the NODE architecture with added replicator-style ecological dynamics:

$$\frac{dy_i}{dt} = y_i(f_i - \bar{f})$$

where y_i represents the abundance of species i , f_i represents its learned interaction dynamics, and $\bar{f}$ is the average community interaction term. This cNODE2 architecture adds a biologically motivated inductive bias in which microbial abundances evolve through competitive compositional dynamics.

Table 1: Inductive biases imposed by each predictive architecture.

Model	Inductive Bias
Linear	Assumes microbial abundance relationships can be approximated through linear interactions between species presence and abundance.
Residual MLP	Assumes complex nonlinear relationships can be learned through deep feedforward transformations and hierarchical feature representations.
Attention	Assumes species importance depends on contextual interactions with other present species through learned pairwise attention mechanisms.
NODE	Assumes microbial communities evolve through continuous dynamical system behavior governed by differential equations.
cNODE2	Assumes microbial abundances evolve through competitive compositional dynamics using replicator-style ecological interactions.

3 Methods

We evaluated multiple predictive architectures on the same microbial abundance prediction task and compared the resulting global keystone species rankings across 100 species in a community. The evaluated models included a linear model, a residual multilayer perceptron (Residual MLP), an attention-based model, a Neural Ordinary Differential Equation (NODE) model, and the original cNODE2 architecture proposed by [7].

The dataset consists of microbial relative abundance compositions represented as a species-by-sample matrix P . Binary species presence vectors Z were generated by thresholding abundances such that all nonzero abundances were mapped to 1.0. Both abundance and binary presence vectors were normalized to sum to one across species. The dataset was divided into training, validation, and test sets, where the test set was reserved for keystone perturbation experiments. Additional dataset details are available in the original DKI repository¹.

All models were trained to learn a mapping from binary species presence vectors z to relative abundance compositions p . Training minimized the Bray–Curtis dissimilarity loss:

$$BC(p, q) = \frac{\sum_i |p_i - q_i|}{\sum_i |p_i + q_i|}$$

where p represents the predicted abundance composition and q represents the ground truth abundance composition.

After training, the keystone species identification ‘thought experiment’ was conducted through simulated perturbation experiments. For each microbial community sample, one species was removed from the binary presence vector and the trained model predicted the resulting perturbed abundance composition. Keystoneness was quantified as the Bray–Curtis

1. Data available at <https://github.com/spxuw/DKI>

dissimilarity between the original and perturbed microbial compositions. More information keystone-ness and the calculations are in [7]. Global keystone rankings were then computed by taking the median keystone-ness value for each species across all perturbation samples.

To compare ranking consistency across architectures, we computed pairwise Spearman rank correlations and Top- k Jaccard similarities between the global species rankings generated by each model. Spearman correlation measured agreement across the full ranking structure, while Top- k Jaccard similarity evaluated overlap among the highest-ranked keystone species, which have the highest ecological relevance. Ranking agreement was additionally visualized using correlation matrices and pairwise scatter plots.

4 Implementation Details

All models were implemented in PyTorch and optimized using the Adam optimizer. Architecture-specific learning rates and training epochs were selected empirically to achieve stable convergence and comparable predictive performance, as shown in Table 2. A batch size of 256 was used for all experiments. Training and validation losses were recorded throughout optimization, and the checkpoint with the lowest validation loss was retained for evaluation.

Table 2: Training hyperparameters for each predictive architecture.

Model	Epochs	Learning Rate
cNODE2	300	0.01
NODE	1000	0.01
Linear	5000	0.10
Residual MLP	5000	0.01
Attention	5000	0.01

The linear, residual MLP, and attention-based architectures predicted abundance compositions directly through feedforward transformations followed by softmax normalization to ensure the relative abundance vector lies on the simplex.

To improve computational efficiency, the cNODE2 implementation used explicit fixed-step Euler integration over a fixed trajectory length rather than a black-box ODE solver, reproducing the original DKI training procedure while reducing runtime. Abundance vectors were clamped to remain nonnegative and renormalized after each integration step.

The NODE architecture used equilibrium-based integration in which the dynamical system was iteratively evolved until convergence. Convergence was determined by observing the change between successive abundance vectors and terminating integration once the change fell below a tolerance threshold. An equilibrium regularization term was included in the loss function during training for steady-state behavior.

Experiments were conducted in Python using PyTorch, NumPy, and Matplotlib. Training and evaluation were performed on GPU-enabled Google Colab instances using CUDA acceleration.

5 Results

Table 3 summarizes the validation Bray–Curtis dissimilarity achieved by each predictive architecture. The NODE and cNODE2 models achieved the lowest validation losses, indicating that continuous dynamical systems were most effective at predicting microbial abundance compositions. This was an expected result as the underlying ecological dynamics are continuous physical systems. In contrast, the linear, attention-based, and residual MLP models achieved higher but still relatively low validation losses.

Table 3: Validation Bray–Curtis loss for each predictive architecture.

Model	Validation Loss
cNODE2	0.00631
NODE	0.00630
Linear	0.0230
Attention	0.0220
Residual MLP	0.0272

Despite differences in predictive performance, all architectures produced highly consistent global keystone species rankings across the 100 species. Figure 1 shows the pairwise Spearman correlation matrix between the global keystone rankings generated by each model. All pairwise comparisons produced Spearman correlations greater than 0.99, indicating extremely strong agreement in the relative ordering of keystone species across architectures. Top-10 Jaccard similarities showed high overlap among the highest-ranked keystone species as shown in Figure 2.

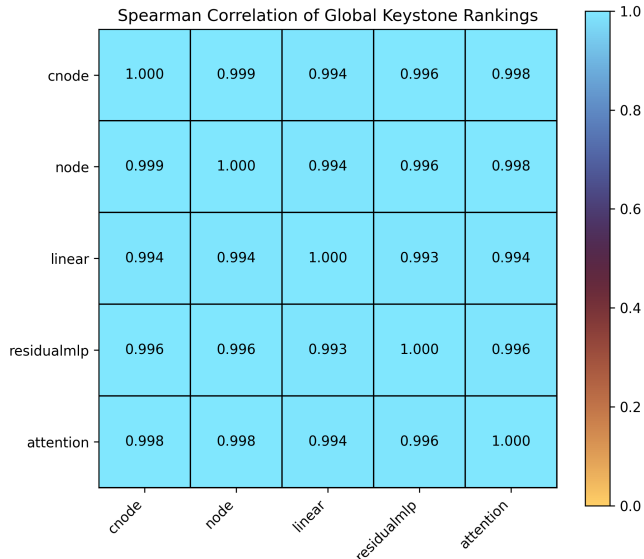


Figure 1: Pairwise Spearman correlation matrix comparing global keystone species rankings across each architecture.

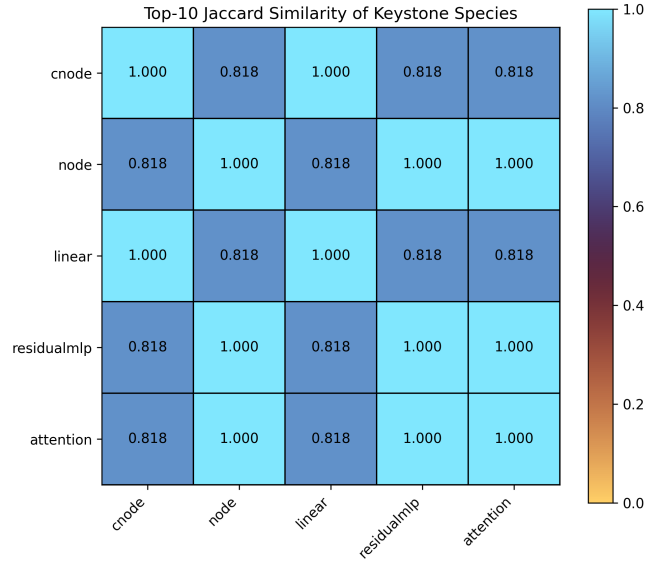


Figure 2: Top-10 Jaccard similarity matrix showing overlap among the highest-ranked keystone species identified by each architecture.

Figure 3 shows pairwise scatter plots comparing the median keystone-ness assigned to each species by cNODE2 and several alternative architectures. Each point represents a microbial species, while the diagonal line indicates identical keystone-ness predictions between models.

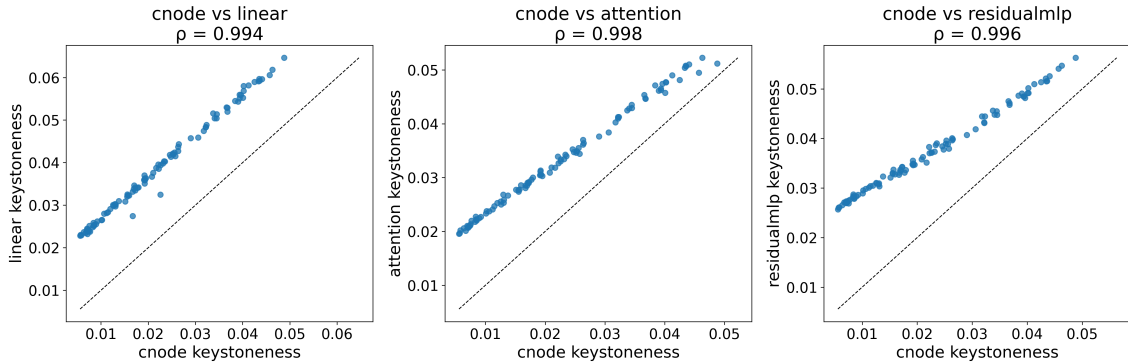


Figure 3: Pairwise scatter plots comparing median keystone-ness assigned to each species by cNODE2 and alternative predictive architectures. Points near the diagonal indicate strong agreement between models.

An important thing to note here is that there are noticeable differences in perturbation magnitude, as indicated by the left shift from the $y=x$ line. In general, the non-ODE

architectures produced larger keystone magnitudes than the NODE models. A possible explanation is that the softmax-based prediction models (linear, residual MLP, and attention) amplify perturbation abundance shifts through exponential normalization, while the ODE-based models evolve compositions gradually through continuous replicator-style dynamics. Consequently, the models appear to agree strongly on which species are influential within the microbial community, while differing in the predicted severity of ecosystem response following species removal.

6 Discussion

The primary finding of this work is that microbial keystone species rankings remained highly consistent across substantially different predictive architectures. Despite the models having different inductive biases and achieving different predictive losses, the resulting global keystone rankings exhibited extremely high Spearman correlations and strong Top- k Jaccard overlap. These results suggest that the DKI framework may identify biologically meaningful keystone species rather than artifacts of a specific model architecture.

Although the ranking structure remained highly consistent across architectures, systematic differences in perturbation magnitude were observed. The non-ODE models generally produced larger keystone values than the NODE and cNODE2 models, possibly due to softmax normalization amplifying abundance shifts, while the ODE-based models evolved compositions more gradually through continuous dynamics. As a result, the models largely agreed on which species were important while differing in the predicted severity of ecosystem response following species removal.

The linear model achieved surprisingly competitive performance relative to the more complex architectures. This suggests that the abundance prediction task may be reasonably approximated by a simple linear mapping within this limited dataset.

These results have important implications for the interpretability and reproducibility of data-driven microbiome models. Since ground truth keystone labels are generally unavailable, robustness across architectures provides an important indicator of trustworthiness. The strong agreement observed across models suggests that the DKI framework may produce stable keystone rankings despite different predictive assumptions, although perturbation magnitudes remain architecture-dependent.

7 Conclusion

In this work, the robustness of microbial keystone species identification within the DKI framework across multiple predictive architectures were evaluated, including linear, attention-based, residual MLP, NODE, and cNODE2 models.

Keystone rankings remained highly consistent across all evaluated architectures, with pairwise Spearman correlations exceeding 0.99 in nearly all comparisons. However, predictive performance and perturbation magnitudes remained architecture-dependent, with ODE-based models achieving lower Bray-Curtis loss and generally producing smaller perturbation magnitudes than direct prediction architectures.

The relatively small keystone magnitudes observed in this study are consistent with the original DKI framework, which reported that structural keystone values in real mi-

crobial communities are typically far below the theoretical upper bound due to the high functional redundancy present in complex microbiomes [7]. Consequently, the relative ranking of species may be more informative than the absolute keystone magnitude itself.

Overall, these results suggest that the DKI framework may identify biologically meaningful keystone species rather than artifacts of a specific model architecture, extending the original observations of Wang et al. [7].

Code Availability

The code used for model training, keystone species evaluation, and figure generation is publicly available at:

https://github.com/williamlittle423/DKI_InductiveBias

References

- [1] J. R. Bray and J. T. Curtis. An ordination of the upland forest communities of southern wisconsin. *Ecological Monographs*, 27(4):325–349, 1957.
- [2] R. T. Q. Chen, Y. Rubanova, J. Bettencourt, and D. Duvenaud. Neural ordinary differential equations. *arXiv preprint arXiv:1806.07366*, 2018. doi: 10.48550/arXiv.1806.07366. URL <https://doi.org/10.48550/arXiv.1806.07366>.
- [3] I. Goodfellow, Y. Bengio, and A. Courville. *Deep Learning*. MIT Press, 2016. URL <https://www.deeplearningbook.org>.
- [4] K. He, X. Zhang, S. Ren, and J. Sun. Deep residual learning for image recognition. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pages 770–778, 2016.
- [5] R. T. Paine. A note on trophic complexity and community stability. *The American Naturalist*, 103(929):91–93, 1969.
- [6] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, and I. Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems*, volume 30, 2017.
- [7] X.-W. Wang, Z. Sun, H. Jia, S. Michel-Mata, M. T. Angulo, L. Dai, X. He, S. T. Weiss, and Y.-Y. Liu. Identifying keystone species in microbial communities using deep learning. *Nature Communications*, 2024. doi: 10.1038/s41559-023-02250-2.